5.13.48

EE25BTECH11022 - sankeerthan

Question: If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of equal degree, then which one is correct among the followings?

1) $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
2) $\mathbf{A} + \mathbf{B} = \mathbf{A} - \mathbf{B}$
3) $\mathbf{A} - \mathbf{B} = \mathbf{B} - \mathbf{A}$
4) $\mathbf{AB} = \mathbf{BA}$

Solution: For any two matrices $\mathbf{A}$ and $\mathbf{B}$ of the same order, addition is commutative Therefore, $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.
let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 3 & 6 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 5 & 3 \\ 8 & 2 \end{pmatrix} \tag{1}$$

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 2+5 & 3+1 \\ 3+8 & 6+2 \end{pmatrix} \tag{2}$$

$$= \begin{pmatrix} 7 & 4 \\ 11 & 8 \end{pmatrix} \tag{3}$$

$$\mathbf{B} + \mathbf{A} = \begin{pmatrix} 5+2 & 1+3 \\ 8+3 & 2+6 \end{pmatrix} \tag{4}$$

$$= \begin{pmatrix} 7 & 4 \\ 11 & 8 \end{pmatrix} \tag{5}$$

Therefore, $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.